



SCHOOL HOMEWORK - 2023

CLASS: INCIDENT RESPONSE AND
DETECTION

SUBJECT: MITRE ATT&CK — PURPLE TEAMING WITH ATOMIC RED
TEAM TESTS — KIBANA & ALERTING

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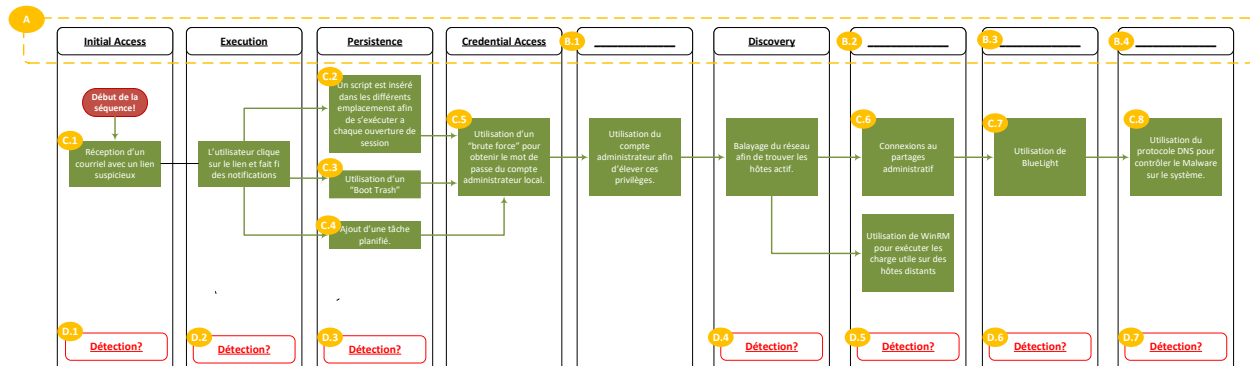
Guideline

- All answers must be documented and supported with screenshots. Even if the answer is correct, if it is not documented **your grade will be 0.**
- You must demonstrate analysis and research in the answers you provide.
- Please submit your assignments as PDF or DOC files and present your answers under the questions below by copying them exactly as written with their question number.
- **Warning:** use your own user account to perform operations on the Security Onion platform.
- All rules that you create for exercises 7 and 8 must contain your first and last name.

Exercise 1 - MITRE ATT&CK (15p)

The following figure represents the modeling of a fictional malware using the MITRE ATT&CK framework. Analyze it and answer the different questions.

To answer each question, refer to the corresponding letter in the figure.



A) In the figure, what do the titles of the box correspond to? (1p)

The tactics

B) Complete what is missing : (2p)

B.1 : Privilege Escalation

B.2 : Lateral Movement

B.3 : Collection

B.4 : Command and Control

C) Identify the corresponding techniques of the MITRE ATT&CK : (4p)

C.1 : Phishing

C.2 : Boot or Logon Autostart Execution

C.3 : Pre-OS Boot

C.4 : Scheduled Task/Job

C.5 : Brute Force

C.6 : Remote Services

C.7 : Archive Collected Data

C.8 : Application Layer Protocol

D) Identify how each of the techniques mentioned could have been detected in the context of the attack (tool / example, etc.) (8p)

D1

Monitor newly created files originating from phishing messages to gain access to victim systems. We could also monitor logs from third-party applications, messages, and/or other artifacts that may send phishing messages to access victim systems. Filtering based on DKIM+SPF or header analysis can help detect when the email sender is spoofed. Inspecting URLs in emails (including expanding shortened links) can help detect links leading to known malicious sites. Sandboxes can be used to detect these links and either automatically visit these sites to determine if they are potentially malicious or wait and capture the content if a user visits the link.

D2

Implement application control, as it may be able to prevent the execution of executables masquerading as other file types. Also, if a link is visited by a user, network intrusion prevention systems such as Suricata, Snort, etc., and systems designed to analyze and remove malicious downloads can be used to block the activity. Finally, if a link is visited by a user, block by default unknown or unused files in transit that should not be downloaded or enforce a policy from suspicious sites as a best practice to prevent

certain vectors such as .scr, .exe, .pif, .cpl, etc. Some download inspection devices can open and analyze compressed and encrypted formats such as ZIP and RAR, which may be used to conceal malicious files.

D3

Monitor executed commands and arguments that may configure system settings to automatically run a program at system startup or user login in order to maintain persistence or gain higher-level privileges on compromised systems. We could also monitor any unusual kernel driver installation activity that may configure system settings to automatically run a program at startup or login to maintain persistence or gain elevated privileges on compromised systems. Also, configure scheduled task settings to force tasks to run under the authenticated user context rather than allowing them to run as SYSTEM. The associated registry key is HKLM\SYSTEM\CurrentControlSet\Control\Lsa\SubmitControl. The setting can be configured via GPO: Computer Configuration > [Policies] > Windows Settings > Security Settings > Security Options > Domain controller: Allow server operators to schedule tasks, set to disabled.

D4

To prevent the technique related to Discovery, ensure that unnecessary ports and services are closed to prevent any risk of discovery and potential exploitation. Secondly, use network intrusion detection/prevention systems to detect and prevent remote service scans. Finally, ensure proper network segmentation is implemented to protect critical servers and devices. For detection, monitor executed commands and arguments that attempt to obtain a list of running services on remote hosts, including those that may be vulnerable to remote software exploitation.

D5

Use multi-factor authentication for remote service logins where possible. Also, limit the accounts that can use remote services. Restrict permissions for accounts that are more likely to be compromised; for example, configure SSH so that users can only execute specific programs. For detection, monitor user accounts logging into systems they would not normally access or exhibiting abnormal access patterns, such as multiple systems within a relatively short time period. Correlate login activity related to remote

services with unusual behavior or other malicious or suspicious activity. Adversaries will likely need to learn the environment and relationships between systems through discovery techniques before attempting lateral movement. For example, on macOS, you can review logs for “screensharing” and “Authentication” event messages.

D6

For detection, monitor executed commands and arguments for actions that facilitate compression or encryption of collected data prior to exfiltration, such as tar. Also, monitor newly created files written with extensions and/or headers associated with compressed or encrypted file types. Detection efforts can focus on subsequent exfiltration activity, where compressed or encrypted files may be detected in transit using a network intrusion detection system or data loss prevention system analyzing file headers. Finally, monitor newly created processes and/or command lines that facilitate compression or encryption of collected data prior to exfiltration, such as 7-Zip, WinRAR, and WinZip.

D7

Monitor and analyze traffic patterns and packet inspection associated with protocols, using SSL/TLS inspection for encrypted traffic, that do not follow protocol standards and expected traffic flows (e.g., superfluous packets that do not belong to established flows, anomalous or unsolicited traffic patterns, abnormal syntax or structure). Consider correlating with process and command-line monitoring to detect execution of abnormal processes and command-line arguments associated with traffic patterns (e.g., monitoring anomalies in the use of files that do not normally initiate connections for the respective protocols). Also, monitor and analyze traffic flows that do not follow protocol standards and expected traffic patterns (e.g., superfluous packets not belonging to established flows or anomalous/unsolicited traffic patterns). Consider correlating with process and command-line monitoring to detect execution of abnormal processes and associated command-line arguments (e.g., monitoring anomalies in the use of files that do not normally initiate connections for the respective protocols).

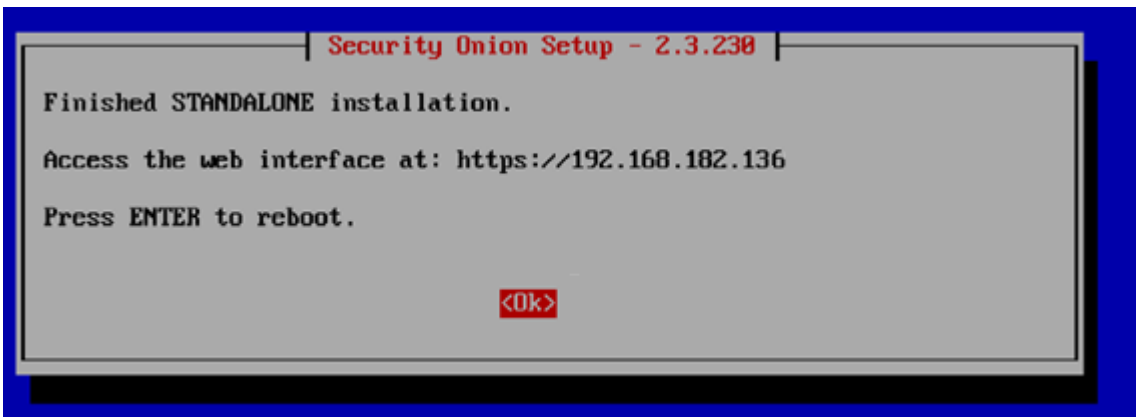
Exercise 2 – Purple Teaming (55p)

For this exercise, you will use WinADHD and Security Onion configured in Standalone mode.

- a) Configure Security Onion in Standalone mode and install Sysmon on WinADHD, then send Sysmon logs to Security Onion. You do not need to document all steps—only a screenshot of your Security Onion Hunt page where we can see Sysmon topics is sufficient (7p)
 - a. If you want, you can use the CentOS version of Security Onion
<https://docs.securityonion.net/en/2.3/installation.html>

Answer

We start by installing Security Onion in Standalone mode.



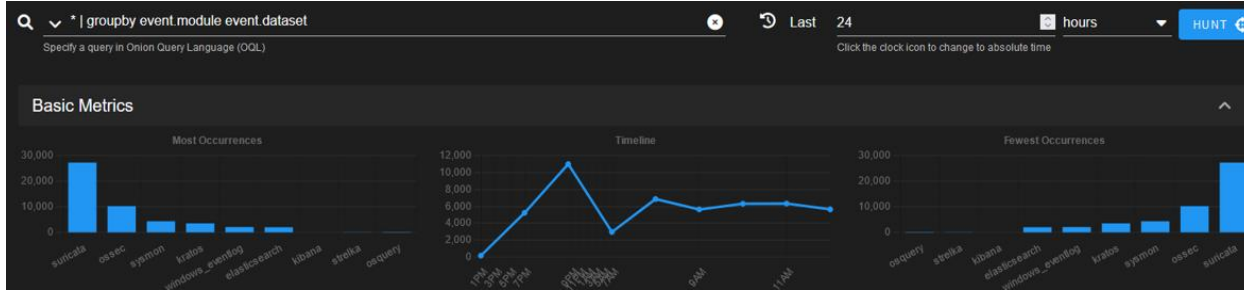
Sysmon installation and configuration

```
PS C:\Users\adhd\Desktop\sysmon> .\Sysmon64.exe -accepteula -i sysmonconfig-export.xml

System Monitor v14.16 - System activity monitor
By Mark Russinovich and Thomas Garnier
Copyright (C) 2014-2023 Microsoft Corporation
Using libxml2. libxml2 is Copyright (C) 1998-2012 Daniel Veillard. All Rights Reserved.
Sysinternals - www.sysinternals.com

Loading configuration file with schema version 4.50
Sysmon schema version: 4.83
Configuration file validated.
Sysmon64 installed.
SysmonDrv installed.
Starting SysmonDrv.
SysmonDrv started.
Starting Sysmon64..
Sysmon64 started.
```

Sysmon information in Security Onion Hunt



- b) Update your Atomic Red Team on the ADHD machine. Execute the following tests and identify them in Security Onion (Hunt or Kibana).
- T1059.005 - Command and Scripting Interpreter: Visual Basic – 1 test (3p)

Here we can see where we update *Atomic Red Team*.

```
PS C:\AtomicRedTeam> IEX (IRM https://raw.githubusercontent.com/redcanaryco/Invoke-AtomicRedTeam/master/install-atomicredteam.ps1 -UseBasicParsing);
PS C:\AtomicRedTeam> Install-AtomicRedTeam -getAtomics -Force
Installation of Invoke-AtomicRedTeam is complete. You can now use the Invoke-AtomicTest Function
See Wiki at https://github.com/redcanaryco/Invoke-AtomicRedTeam/wiki for complete details
```

We execute the command to view brief details of the test in question.

```
Administrator: Windows PowerShell
PS C:\AtomicRedTeam> Invoke-AtomicTest T1059.005 -ShowDetailsBrief
PathToAtomicsFolder = C:\AtomicRedTeam\atomics

T1059.005-1 Visual Basic script execution to gather local computer information
T1059.005-2 Encoded VBS code execution
T1059.005-3 Extract Memory via VBA
PS C:\AtomicRedTeam>
```

We then review the prerequisites, and when there are none, we simply start the test.

```
PS C:\AtomicRedTeam> Invoke-AtomicTest T1059.005 -TestNumbers 1 -CheckPrereqs
PathToAtomicsFolder = C:\AtomicRedTeam\atomics

CheckPrereq's for: T1059.005-1 Visual Basic script execution to gather local computer information
Prerequisites met: T1059.005-1 Visual Basic script execution to gather local computer information
PS C:\AtomicRedTeam>
```

```
PS C:\AtomicRedTeam> Invoke-AtomicTest T1059.005 -TestNumbers 1
PathToAtomicsFolder = C:\AtomicRedTeam\atomics

Executing test: T1059.005-1 Visual Basic script execution to gather local computer information
Done executing test: T1059.005-1 Visual Basic script execution to gather local computer information
PS C:\AtomicRedTeam>
```

Results in Hunt


```

ProcessName:
RuleName: -
UtcTime: 2023-04-24 18:21:56.623
ProcessGuid: {b1a62ae4-c8c4-6446-d205-000000003500}
ProcessId: 8068
Image: C:\Windows\System32\cmd.exe
FileVersion: 5.812.10240.16384
Description: Microsoft® Console Based Script Host
Product: Microsoft® Windows Script Host
Company: Microsoft Corporation
OriginalFileName: cmd.exe
CommandLine: "C:\WINDOWS\system32\cmd.exe" C:\AtomicRedTeam\atomics\T1059.005\src\sys_info.vbs
CurrentDirectory: C:\Users\adhd\AppData\Local\Temp\
User: DESKTOP-IT2G0T\adhd
LogonGuid: {b1a62ae4-d332-6445-204c-060000000000}
LogonId: 0x64C20
TerminalSessionId: 1
IntegrityLevel: High
Hashes: MD5=24590BF74B8B8FD7D7AC070F4E9C44FD.SHA256=AE37FD18642E797B36B9FFCEC8A6E986732D011581061800C6B74426C26A9D03.MP
ParentProcessGuid: {b1a62ae4-c8c4-6446-d205-000000003500}
ParentProcessId: 3068
ParentImage: C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
ParentCommandLine: "powershell.exe" & (cmd C:\AtomicRedTeam\atomics\T1059.005\src\sys_info.vbs > $env:TEMP\T1059.005.out.txt)
ProcessName: DESKTOP-IT2G0T\adhd

```

b. T1053.005 - Scheduled Task/Job: Scheduled Task, 4 tests de votre choix (12p)

Here we can see the different available tests to perform specific actions.

```

T1053.005-5 Extract Memory via VBA
PS C:\AtomicRedTeam> Invoke-AtomicTest T1053.005 -ShowDetailsBrief
PathToAtomicsFolder = C:\AtomicRedTeam\atomics

T1053.005-1 Scheduled Task Startup Script
T1053.005-2 Scheduled task Local
T1053.005-3 Scheduled task Remote
T1053.005-4 Powershell Cmdlet Scheduled Task
T1053.005-5 Task Scheduler via VBA
T1053.005-6 WMI Invoke-CimMethod Scheduled Task
T1053.005-7 Scheduled Task Executing Base64 Encoded Commands From Registry
T1053.005-8 Import XML Schedule Task with Hidden Attribute
T1053.005-9 PowerShell Modify A Scheduled Task
PS C:\AtomicRedTeam>

```

Next, as with the previous test, we check the prerequisites to ensure we are able to execute them.

```

T1053.005-9 PowerShell Modify A Scheduled Task
PS C:\AtomicRedTeam> invoke-atomictest t1053.005 -TestNumbers 1 -CheckPrereqs
PathToAtomicsFolder = C:\AtomicRedTeam\atomics

CheckPrereq's for: T1053.005-1 Scheduled Task Startup Script
Prerequisites met: T1053.005-1 Scheduled Task Startup Script
PS C:\AtomicRedTeam> invoke-atomictest t1053.005 -TestNumbers 2 -CheckPrereqs
PathToAtomicsFolder = C:\AtomicRedTeam\atomics

CheckPrereq's for: T1053.005-2 Scheduled task Local
Prerequisites met: T1053.005-2 Scheduled task Local
PS C:\AtomicRedTeam> invoke-atomictest t1053.005 -TestNumbers 3 -CheckPrereqs
PathToAtomicsFolder = C:\AtomicRedTeam\atomics

CheckPrereq's for: T1053.005-3 Scheduled task Remote
Prerequisites met: T1053.005-3 Scheduled task Remote
PS C:\AtomicRedTeam> invoke-atomictest t1053.005 -TestNumbers 4 -CheckPrereqs
PathToAtomicsFolder = C:\AtomicRedTeam\atomics

CheckPrereq's for: T1053.005-4 Powershell Cmdlet Scheduled Task
Prerequisites met: T1053.005-4 Powershell Cmdlet Scheduled Task

```

Results after executing all of them + the next screenshot will show the results in *Onion*.

```

CheckPrereq's For: T1053.005-4 Powershell Cmdlet Scheduled Task
Prerequisites met: T1053.005-4 Powershell Cmdlet Scheduled Task
PS C:\AtomicRedTeam> Invoke-atomictest t1053.005 -TestNumbers 1
PathToAtomicsFolder = C:\AtomicRedTeam\atomics

Executing test: T1053.005-1 Scheduled Task Startup Script
SUCCESS: The scheduled task "T1053_005_OnLogon" has successfully been created.
SUCCESS: The scheduled task "T1053_005_OnStartup" has successfully been created.
Done executing test: T1053.005-1 Scheduled Task Startup Script
PS C:\AtomicRedTeam> Invoke-atomictest t1053.005 -TestNumbers 2
PathToAtomicsFolder = C:\AtomicRedTeam\atomics

Executing test: T1053.005-2 Scheduled task Local
SUCCESS: The scheduled task "spawn" has successfully been created.
Done executing test: T1053.005-2 Scheduled task Local
PS C:\AtomicRedTeam> Invoke-atomictest t1053.005 -TestNumbers 3
PathToAtomicsFolder = C:\AtomicRedTeam\atomics

Executing test: T1053.005-3 Scheduled task Remote
ERROR: No mapping between account names and security IDs was done.
Done executing test: T1053.005-3 Scheduled task Remote
PS C:\AtomicRedTeam> Invoke-atomictest t1053.005 -TestNumbers 4
PathToAtomicsFolder = C:\AtomicRedTeam\atomics

Executing test: T1053.005-4 Powershell Cmdlet Scheduled Task
TaskPath          TaskName          State
-----
\                  AtomicTask        Ready
Done executing test: T1053.005-4 Powershell Cmdlet Scheduled Task
PS C:\AtomicRedTeam>

```

```

672 -06:00 schtasks /create /sc ONCE /tn spawn /tr C:\windows\system32\cmd.exe /ST 20:10
650 -06:00 "cmd.exe" /c "schtasks /create /sc ONCE /tn spawn /tr C:\windows\system32\cmd.exe /ST 20:10"
501 -06:00 "C:\WINDOWS\system32\whoami.exe"
487 -06:00 "C:\WINDOWS\system32\HOSTNAME.EXE"
545 -06:00 schtasks /create /tn "T1053_005_OnStartup" /sc onstart /ru system /tr "cmd.exe /c calc.exe"
527 -06:00 schtasks /create /tn "T1053_005_OnLogon" /sc onlogon /tr "cmd.exe /c calc.exe"
503 -06:00 "cmd.exe" /c "schtasks /create /tn "T1053_005_OnLogon" /sc onlogon /tr "cmd.exe /c calc.exe" & schtasks /create /tn "T1053_005_O

```

c. T1569.002 - System Services: Service Execution, 4 tests of your choice (12p)

Once again, we start by requesting the details related to the tests we want to execute.

```

T1569.002-1 Powershell Cmdlet Scheduled Task
PS C:\AtomicRedTeam> Invoke-AtomicTest T1569.002 -ShowDetailsBrief
PathToAtomicsFolder = C:\AtomicRedTeam\atomics

T1569.002-1 Execute a Command as a Service
T1569.002-2 Use PsExec to execute a command on a remote host
T1569.002-4 BlackCat pre-encryption cmds with Lateral Movement
T1569.002-5 Use RemCom to execute a command on a remote host
PS C:\AtomicRedTeam>

```

For the first time, we have prerequisites to install for these tests.

```

PS C:\AtomicRedTeam> invoke-atomictest t1569.002 -TestNumbers 1 -CheckPrereqs
PathToAtomicsFolder = C:\AtomicRedTeam\atomics

CheckPrereq's for: T1569.002-1 Execute a Command as a Service
Prerequisites met: T1569.002-1 Execute a Command as a Service
PS C:\AtomicRedTeam> invoke-atomictest t1569.002 -TestNumbers 2 -CheckPrereqs
PathToAtomicsFolder = C:\AtomicRedTeam\atomics

CheckPrereq's for: T1569.002-2 Use PsExec to execute a command on a remote host
Prerequisites not met: T1569.002-2 Use PsExec to execute a command on a remote host
[*] PsExec tool from Sysinternals must exist on disk at specified location (C:\PSTools\PsExec.exe)

Try installing prereq's with the -GetPrereqs switch
PS C:\AtomicRedTeam> invoke-atomictest t1569.002 -TestNumbers 4 -CheckPrereqs
PathToAtomicsFolder = C:\AtomicRedTeam\atomics

CheckPrereq's for: T1569.002-4 BlackCat pre-encryption cmds with Lateral Movement
Prerequisites met: T1569.002-4 BlackCat pre-encryption cmds with Lateral Movement
PS C:\AtomicRedTeam> invoke-atomictest t1569.002 -TestNumbers 5 -CheckPrereqs
PathToAtomicsFolder = C:\AtomicRedTeam\atomics

CheckPrereq's for: T1569.002-5 Use RemCom to execute a command on a remote host
Prerequisites not met: T1569.002-5 Use RemCom to execute a command on a remote host
[*] RemCom tool must exist on disk at specified location (C:\AtomicRedTeam\atomics\T1569.002\bin\remcom.exe)

Try installing prereq's with the -GetPrereqs switch
PS C:\AtomicRedTeam>

```

We can see where we acquire the prerequisites here for tests number 5 and 2.

```

PS C:\AtomicRedTeam> invoke-atomictest t1569.002 -TestNumbers 5 -GetPrereqs
PathToAtomicsFolder = C:\AtomicRedTeam\atomics

GetPrereq's for: T1569.002-5 Use RemCom to execute a command on a remote host
Attempting to satisfy prereq: RemCom tool must exist on disk at specified location (C:\AtomicRedTeam\atomics\T1569.002\bin\remcom.exe)
Prereq successfully met: RemCom tool must exist on disk at specified location (C:\AtomicRedTeam\atomics\T1569.002\bin\remcom.exe)
PS C:\AtomicRedTeam> invoke-atomictest t1569.002 -TestNumbers 2 -GetPrereqs
PathToAtomicsFolder = C:\AtomicRedTeam\atomics

GetPrereq's for: T1569.002-2 Use PsExec to execute a command on a remote host
Attempting to satisfy prereq: PsExec tool from Sysinternals must exist on disk at specified location (C:\PSTools\PsExec.exe)
Prereq successfully met: PsExec tool from Sysinternals must exist on disk at specified location (C:\PSTools\PsExec.exe)
PS C:\AtomicRedTeam>

```

Here we can see that the prerequisites have now been obtained.

```

PS C:\AtomicRedTeam> invoke-atomictest t1569.002 -TestNumbers 2 -CheckPrereqs
PathToAtomicsFolder = C:\AtomicRedTeam\atomics

CheckPrereq's for: T1569.002-2 Use PsExec to execute a command on a remote host
Prerequisites met: T1569.002-2 Use PsExec to execute a command on a remote host
PS C:\AtomicRedTeam> invoke-atomictest t1569.002 -TestNumbers 5 -CheckPrereqs
PathToAtomicsFolder = C:\AtomicRedTeam\atomics

CheckPrereq's for: T1569.002-5 Use RemCom to execute a command on a remote host
Prerequisites met: T1569.002-5 Use RemCom to execute a command on a remote host
PS C:\AtomicRedTeam>

```

And the results when we execute the tests.

```

PS C:\AtomicRedTeam> invoke-atomictest t1569.002 -TestNumbers 1
PathToAtomsFolder = C:\AtomicRedTeam\atomsics

The service did not respond
StartService FAILED 1053:

PS C:\AtomicRedTeam> invoke-atomictest t1569.002 -TestNumbers 2
PathToAtomsFolder = C:\AtomicRedTeam\atomsics

Executing test: T1569.002-2 Use PsExec to execute a command on a remote host
PsExec v2.43 - Execute processes remotely
Copyright (C) 2001-2023 Mark Russinovich
Sysinternals - www.sysinternals.com
Connecting to localhost...
Starting PSEXESVC service on localhost...
Copying authentication key to localhost...
Connecting with PsExec service on localhost...
Starting C:\Windows\System32\calc.exe on localhost...
PsExec could not start C:\Windows\System32\calc.exe on localhost:
The user name or password is incorrect.
Done executing test: T1569.002-2 Use PsExec to execute a command on a remote host
PS C:\AtomicRedTeam> invoke-atomictest t1569.002 -TestNumbers 4
PathToAtomsFolder = C:\AtomicRedTeam\atomsics

Executing test: T1569.002-4 BlackCat pre-encryption cmds with Lateral Movement
UUID
14034D56-80E3-92D6-8F99-DCDAC42F472F
The operation completed successfully.
PsExec v2.4 - Execute processes remotely
Copyright (C) 2001-2022 Mark Russinovich
Sysinternals - www.sysinternals.com
cmd.exe exited with error code 0.
Done executing test: T1569.002-4 BlackCat pre-encryption cmds with Lateral Movement
PS C:\AtomicRedTeam> invoke-atomictest t1569.002 -TestNumbers 5
PathToAtomsFolder = C:\AtomicRedTeam\atomsics

Executing test: T1569.002-5 Use RemCom to execute a command on a remote host
Localhost entered for Target Machine .. Going to RunAs Command
Remote Command Executor
Copyright 2006 The WiseGuyz [ http://talhatariq.wordpress.com ]
Copyright 2012 Telefonica Global Technology
Author: Talha Tariq [talha.tariq@gmail.com]
Contributor: Luke Suchocki
Contributor: Merlyn Morgan-Graha
Contributor: Andres Ederra
Local Admin
Launching Local Process ...
Done executing test: T1569.002-5 Use RemCom to execute a command on a remote host
PS C:\AtomicRedTeam>

```

Now, the results of these in *Onion Hunt*

06:00	C:\Windows\System32\WindowsPowerShell\1.0\powershell.exe	DESKTOP-ITJ3
06:00	C:\Windows\System32\WindowsPowerShell\1.0\powershell.exe	DESKTOP-ITJ3
06:00	C:\Windows\System32\WindowsPowerShell\1.0\powershell.exe	DESKTOP-ITJ3
06:00	C:\Windows\System32\WindowsPowerShell\1.0\powershell.exe	DESKTOP-ITJ3
06:00	C:\Windows\System32\WindowsPowerShell\1.0\powershell.exe	DESKTOP-ITJ3
06:00	C:\Windows\System32\WindowsPowerShell\1.0\powershell.exe	DESKTOP-ITJ3
06:00	C:\Windows\System32\WindowsPowerShell\1.0\powershell.exe	DESKTOP-ITJ3
06:00	C:\Windows\System32\WindowsPowerShell\1.0\powershell.exe	DESKTOP-ITJ3
14:47:21.282 -06:00	C:\AtomicRedTeam\atomsics\T1569.002\bin\remcom.exe	DESKTOP-ITJ3
14:42:52.999 -06:00	C:\AtomicRedTeam\atomsics\T1569.002\bin\remcom.exe	DESKTOP-ITJ3

- d. T1003.001 - OS Credential Dumping: LSASS Memory – 5 tests of your choice (15p)

Once again, we begin by retrieving the details for the requested tests.

```
T1003.002-3 USE REMCOM TO EXECUTE A COMMAND ON A REMOTE HOST
PS C:\AtomicRedTeam> Invoke-AtomicTest T1003.001 -ShowDetailsBrief
PathToAtomicsFolder = C:\AtomicRedTeam\atomics

T1003.001-1 Dump LSASS.exe Memory using ProcDump
T1003.001-2 Dump LSASS.exe Memory using comsvcs.dll
T1003.001-3 Dump LSASS.exe Memory using direct system calls and API unhooking
T1003.001-4 Dump LSASS.exe Memory using NanoDump
T1003.001-6 Offline Credential Theft With Mimikatz
T1003.001-7 LSASS read with pypykatz
T1003.001-8 Dump LSASS.exe Memory using Out-Minidump.ps1
T1003.001-9 Create Mini Dump of LSASS.exe using ProcDump
T1003.001-10 Powershell Mimikatz
T1003.001-11 Dump LSASS with createdump.exe from .Net v5
T1003.001-12 Dump LSASS.exe using imported Microsoft DLLs
PS C:\AtomicRedTeam>
```

This time, we have more than two tests without prerequisites. We need to acquire them.

```
PS C:\AtomicRedTeam> Invoke-AtomicTest t1003.001 -TestNumbers 1 -CheckPrereqs
PathToAtomicsFolder = C:\AtomicRedTeam\atomics

CheckPrereq's for: T1003.001-1 Dump LSASS.exe Memory using ProcDump
Prerequisites not met: T1003.001-1 Dump LSASS.exe Memory using ProcDump
[*] Procdump tool from Sysinternals must exist on disk at specified location (C:\AtomicRedTeam\atomics\T1003.001\bin\

Try installing prereq's with the -GetPrereqs switch
PS C:\AtomicRedTeam> Invoke-AtomicTest t1003.001 -TestNumbers 2 -CheckPrereqs
PathToAtomicsFolder = C:\AtomicRedTeam\atomics

CheckPrereq's for: T1003.001-2 Dump LSASS.exe Memory using comsvcs.dll
Prerequisites met: T1003.001-2 Dump LSASS.exe Memory using comsvcs.dll
PS C:\AtomicRedTeam> Invoke-AtomicTest t1003.001 -TestNumbers 3 -CheckPrereqs
PathToAtomicsFolder = C:\AtomicRedTeam\atomics

CheckPrereq's for: T1003.001-3 Dump LSASS.exe Memory using direct system calls and API unhooking
Prerequisites not met: T1003.001-3 Dump LSASS.exe Memory using direct system calls and API unhooking
[*] Dumpert executable must exist on disk at specified location (C:\AtomicRedTeam\atomics\T1003.001\bin\Outflank-Dump

Try installing prereq's with the -GetPrereqs switch
PS C:\AtomicRedTeam> Invoke-AtomicTest t1003.001 -TestNumbers 4 -CheckPrereqs
PathToAtomicsFolder = C:\AtomicRedTeam\atomics

CheckPrereq's for: T1003.001-4 Dump LSASS.exe Memory using NanoDump
Prerequisites not met: T1003.001-4 Dump LSASS.exe Memory using NanoDump
[*] NanoDump executable must exist on disk at specified location ($env:TEMP\nanodump.x64.exe)

Try installing prereq's with the -GetPrereqs switch
PS C:\AtomicRedTeam> Invoke-AtomicTest t1003.001 -TestNumbers 6 -CheckPrereqs
PathToAtomicsFolder = C:\AtomicRedTeam\atomics

CheckPrereq's for: T1003.001-6 Offline Credential Theft With Mimikatz
Prerequisites not met: T1003.001-6 Offline Credential Theft With Mimikatz
[*] Mimikatz must exist on disk at specified location (C:\AtomicRedTeam\atomics\T1003.001\bin\x64\mimikatz.exe)

Try installing prereq's with the -GetPrereqs switch
PS C:\AtomicRedTeam>
```

We then obtain the required prerequisites.

```
PS C:\AtomicRedTeam> Invoke-AtomicTest t1003.001 -TestNumbers 1 -GetPrereqs
PathToAtomicsFolder = C:\AtomicRedTeam\atomics

GetPrereq's for: T1003.001-1 Dump LSASS.exe Memory using ProcDump
Attempting to satisfy prereq: Procdump tool from Sysinternals must exist on disk at specified location (C:\AtomicRedTeam\atomics\T1003.001\bin\procdump.exe)
Prereq successfully met: Procdump tool from Sysinternals must exist on disk at specified location (C:\AtomicRedTeam\atomics\T1003.001\bin\procdump.exe)
PS C:\AtomicRedTeam> Invoke-AtomicTest t1003.001 -TestNumbers 3 -GetPrereqs
PathToAtomicsFolder = C:\AtomicRedTeam\atomics

GetPrereq's for: T1003.001-3 Dump LSASS.exe Memory using direct system calls and API unhooking
Attempting to satisfy prereq: Dumpert executable must exist on disk at specified location (C:\AtomicRedTeam\atomics\T1003.001\bin\Outflank-Dumpert.exe)
Prereq successfully met: Dumpert executable must exist on disk at specified location (C:\AtomicRedTeam\atomics\T1003.001\bin\Outflank-Dumpert.exe)
PS C:\AtomicRedTeam> Invoke-AtomicTest t1003.001 -TestNumbers 4 -GetPrereqs
PathToAtomicsFolder = C:\AtomicRedTeam\atomics

GetPrereq's for: T1003.001-4 Dump LSASS.exe Memory using NanoDump
Attempting to satisfy prereq: NanoDump executable must exist on disk at specified location ($env:TEMP\nanodump.x64.exe)
Prereq successfully met: NanoDump executable must exist on disk at specified location ($env:TEMP\nanodump.x64.exe)
PS C:\AtomicRedTeam> Invoke-AtomicTest t1003.001 -TestNumbers 6 -GetPrereqs
PathToAtomicsFolder = C:\AtomicRedTeam\atomics

GetPrereq's for: T1003.001-6 Offline Credential Theft With Mimikatz
Attempting to satisfy prereq: Mimikatz must exist on disk at specified location (C:\AtomicRedTeam\atomics\T1003.001\bin\x64\mimikatz.exe)
Prereq successfully met: Mimikatz must exist on disk at specified location (C:\AtomicRedTeam\atomics\T1003.001\bin\x64\mimikatz.exe)
Attempting to satisfy prereq: Lsass dump must exist at specified location (%tmp%\lsass.DMP)
Prereq already met: Lsass dump must exist at specified location (%tmp%\lsass.DMP)
PS C:\AtomicRedTeam>
```

Next, as usual, we review the prerequisites again to ensure they have been properly applied.

```
PS C:\AtomicRedTeam> Invoke-AtomicTest t1003.001 -TestNumbers 1 -CheckPrereqs
PathToAtomicsFolder = C:\AtomicRedTeam\atomics

CheckPrereq's for: T1003.001-1 Dump LSASS.exe Memory using ProcDump
Prerequisites met: T1003.001-1 Dump LSASS.exe Memory using ProcDump
PS C:\AtomicRedTeam> Invoke-AtomicTest t1003.001 -TestNumbers 2 -CheckPrereqs
PathToAtomicsFolder = C:\AtomicRedTeam\atomics

CheckPrereq's for: T1003.001-2 Dump LSASS.exe Memory using comsvcs.dll
Prerequisites met: T1003.001-2 Dump LSASS.exe Memory using comsvcs.dll
PS C:\AtomicRedTeam> Invoke-AtomicTest t1003.001 -TestNumbers 3 -CheckPrereqs
PathToAtomicsFolder = C:\AtomicRedTeam\atomics

CheckPrereq's for: T1003.001-3 Dump LSASS.exe Memory using direct system calls and API unhooking
Prerequisites met: T1003.001-3 Dump LSASS.exe Memory using direct system calls and API unhooking
PS C:\AtomicRedTeam> Invoke-AtomicTest t1003.001 -TestNumbers 4 -CheckPrereqs
PathToAtomicsFolder = C:\AtomicRedTeam\atomics

CheckPrereq's for: T1003.001-4 Dump LSASS.exe Memory using NanoDump
Prerequisites met: T1003.001-4 Dump LSASS.exe Memory using NanoDump
PS C:\AtomicRedTeam> Invoke-AtomicTest t1003.001 -TestNumbers 6 -CheckPrereqs
PathToAtomicsFolder = C:\AtomicRedTeam\atomics

CheckPrereq's for: T1003.001-6 Offline Credential Theft With Mimikatz
Prerequisites met: T1003.001-6 Offline Credential Theft With Mimikatz
PS C:\AtomicRedTeam>
```

After that, we can execute the tests using different commands for tests 1, 2, and 3 (the selected tests are numbers 1, 2, 3, 4, and 6).

```
PS C:\AtomicRedTeam> Invoke-AtomicTest t1003.001 -TestNumbers 1
PathToAtomicsFolder = C:\AtomicRedTeam\atomics

Executing test: T1003.001-1 Dump LSASS.exe Memory using ProcDump
Copyright (C) 2009-2022 Mark Russinovich and Andrew Richards
Sysinternals - www.sysinternals.com
Procdump v11.0 - Sysinternals process dump utility
[15:19:02] Dump 1 initiated: C:\Windows\Temp\lsass_dump.dmp
[15:19:02] Dump 1 writing: Estimated dump file size is 58 MB.
[15:19:03] Dump 1 complete: 59 MB written in 0.8 seconds
[15:19:03] Dump count reached.
Done executing test: T1003.001-1 Dump LSASS.exe Memory using ProcDump
PS C:\AtomicRedTeam> Invoke-AtomicTest t1003.001 -TestNumbers 2
PathToAtomicsFolder = C:\AtomicRedTeam\atomics

Executing test: T1003.001-2 Dump LSASS.exe Memory using comsvcs.dll
Done executing test: T1003.001-2 Dump LSASS.exe Memory using comsvcs.dll
PS C:\AtomicRedTeam> Invoke-AtomicTest t1003.001 -TestNumbers 3
PathToAtomicsFolder = C:\AtomicRedTeam\atomics
```

Then, using different commands, we execute test number 4.

- e. T1136.001 - Create Account: Local Account, 2 tests of your choice (6p)

For the final test, here is the command to view its details.

```
T1136.001-12 Dump LSASS.exe using imported Microsoft DLLs
PS C:\AtomicRedTeam> Invoke-AtomicTest T1136.001 -ShowDetailsBrief
PathToAtomicsFolder = C:\AtomicRedTeam\atomics

T1136.001-3 Create a new user in a command prompt
T1136.001-4 Create a new user in PowerShell
T1136.001-6 Create a new Windows admin user
PS C:\AtomicRedTeam>
```

We check whether we have the required prerequisites for the test.

```
PS C:\AtomicRedTeam> Invoke-AtomicTest T1136.001 -TestNumbers 3 -CheckPrereqs
PathToAtomicsFolder = C:\AtomicRedTeam\atomics

CheckPrereq's for: T1136.001-3 Create a new user in a command prompt
Prerequisites met: T1136.001-3 Create a new user in a command prompt
PS C:\AtomicRedTeam> Invoke-AtomicTest T1136.001 -TestNumbers 4 -CheckPrereqs
PathToAtomicsFolder = C:\AtomicRedTeam\atomics

CheckPrereq's for: T1136.001-4 Create a new user in PowerShell
Prerequisites met: T1136.001-4 Create a new user in PowerShell
PS C:\AtomicRedTeam> Invoke-AtomicTest T1136.001 -TestNumbers 6 -CheckPrereqs
PathToAtomicsFolder = C:\AtomicRedTeam\atomics

CheckPrereq's for: T1136.001-6 Create a new Windows admin user
Prerequisites met: T1136.001-6 Create a new Windows admin user
PS C:\AtomicRedTeam> _
```

Since the prerequisites are already in place, we can simply execute the tests.

```
PS C:\AtomicRedTeam> Invoke-AtomicTest T1136.001 -TestNumbers 3
PathToAtomicsFolder = C:\AtomicRedTeam\atomics

Executing test: T1136.001-3 Create a new user in a command prompt
The command completed successfully.
Done executing test: T1136.001-3 Create a new user in a command prompt
PS C:\AtomicRedTeam> Invoke-AtomicTest T1136.001 -TestNumbers 4
PathToAtomicsFolder = C:\AtomicRedTeam\atomics

Executing test: T1136.001-4 Create a new user in PowerShell
Name                Enabled Description
----                -
T1136.001_PowerShell True
Done executing test: T1136.001-4 Create a new user in PowerShell
PS C:\AtomicRedTeam>
```

Results in Onion

>	2023-04-24 17:03:20.046 -06:00	C:\WINDOWS\system32\net1 localgroup administrators "T1136.001_Admin" /add	7120	C:\Windows\System32\net.exe	C:\Users\adhd\AppData\Local\Temp\
>	2023-04-24 17:03:20.038 -06:00	net localgroup administrators "T1136.001_Admin" /add	10492	C:\Windows\System32\cmd.exe	C:\Users\adhd\AppData\Local\Temp\
>	2023-04-24 17:03:20.046 -06:00	C:\WINDOWS\system32\net1 user /add "T1136.001_Admin" "T1136_pass"	4136	C:\Windows\System32\net.exe	C:\Users\adhd\AppData\Local\Temp\
>	2023-04-24 17:03:20.046 -06:00	net user /add "T1136.001_Admin" "T1136_pass"	9644	C:\Windows\System32\cmd.exe	C:\Users\adhd\AppData\Local\Temp\
>	2023-04-24 17:03:20.046 -06:00	"cmd.exe" /c "net user /add "T1136.001_Admin" "T1136_pass" & net localgroup administrators "T1136.001_Admin" /add"	2060	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	C:\Users\adhd\AppData\Local\Temp\

Exercise 3 – Kibana & Alerting (30p)

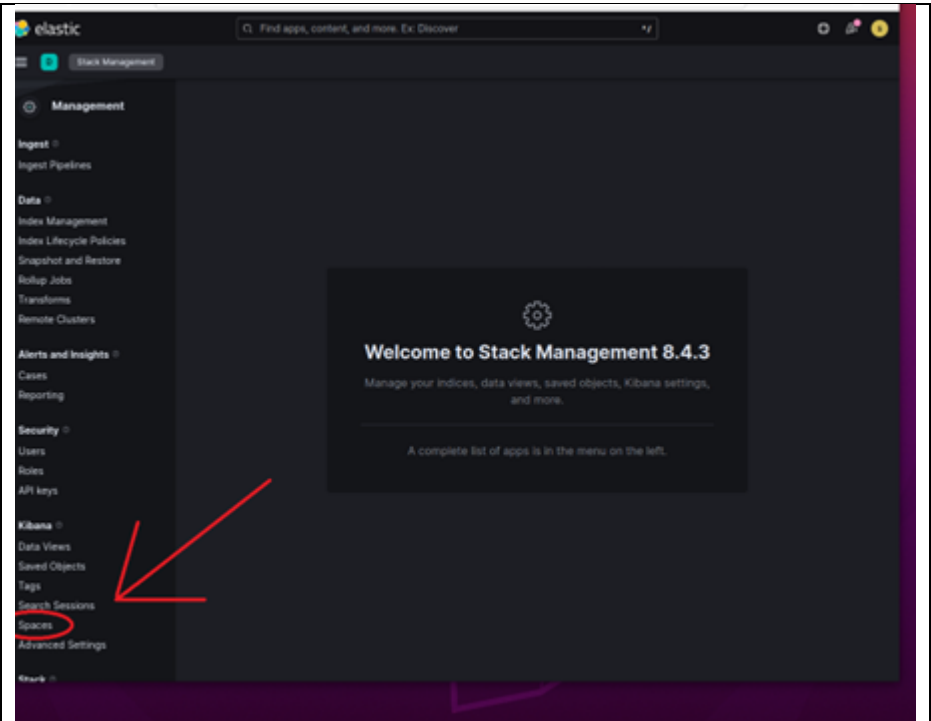
- a) Complete the labs from course 12, using usernames related to your first or last name. (10p)

For each rule, include a screenshot :

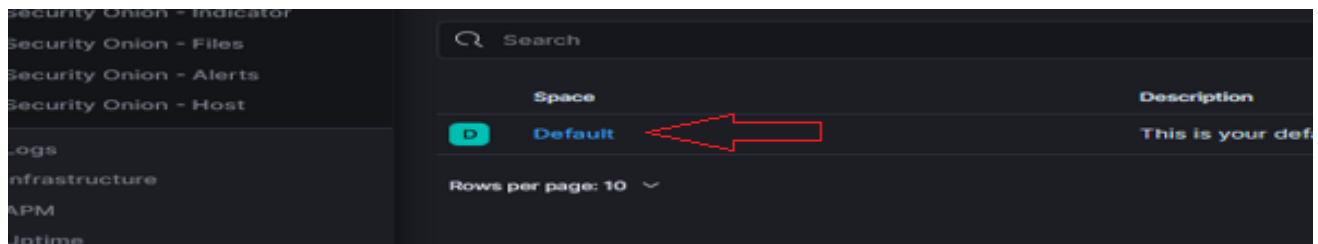
- Of the rule
- Of the base event (used to trigger the rule)
- Of the correlated event

Answer

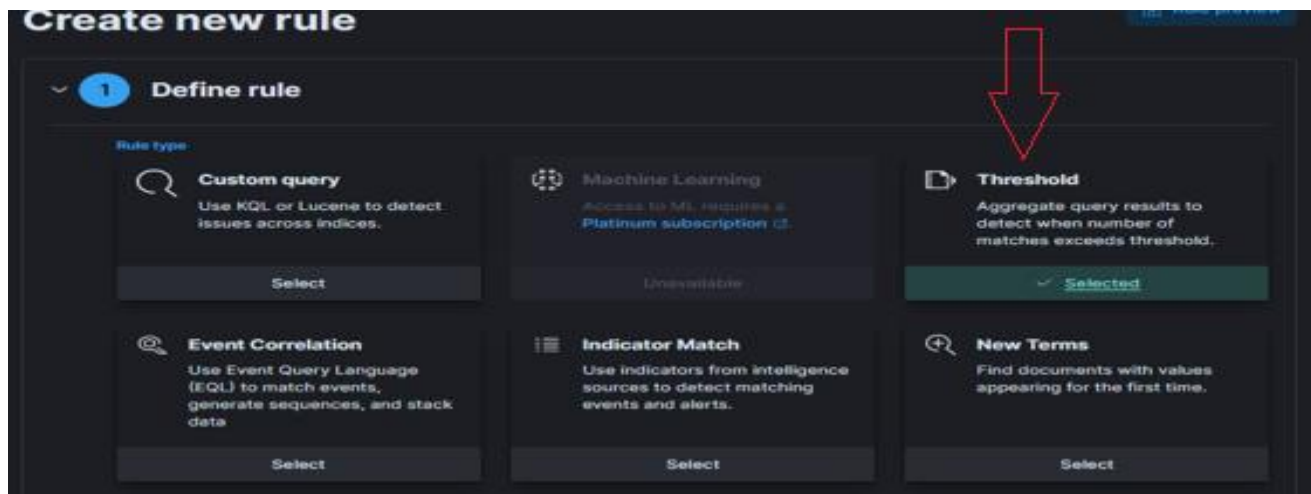
We start by locating “Spaces” under Kibana.



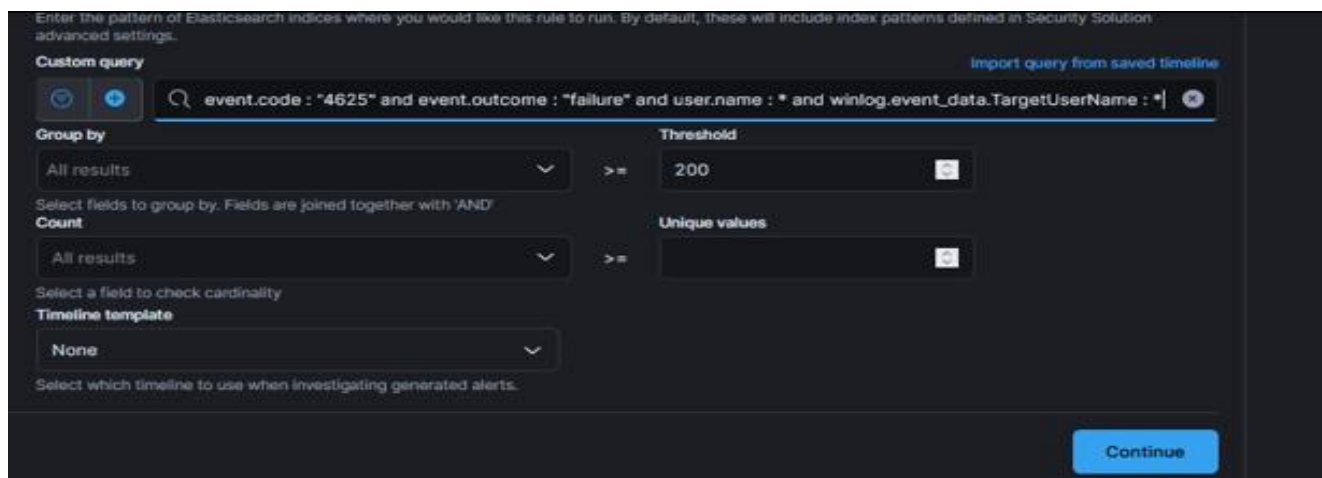
After, we head to Defaults



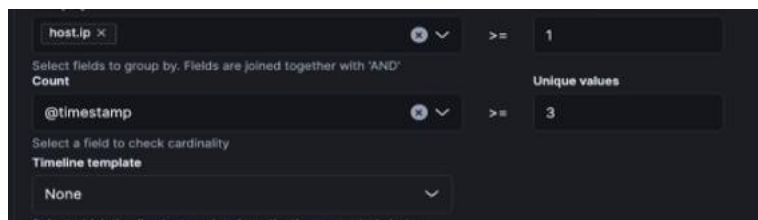
We create our own rule with section *Threshold*



Our rule



We must then modify the group and count



We can then create the rule

2

About rule

Name

CR450-Frederic-Perron

Description

450

Default severity

Select a severity level for all alerts generated by this rule.

Low

☐ Severity override

Use source event values to override the default severity.

Default risk score

Select a risk score for all alerts generated by this rule.

0255075100

21

☐ Risk score override

Use a source event value to override the default risk score.

Tags

Optional

Type one or more custom identifying tags for this rule. Press enter after each tag to begin a new one.

We set the schedule of the rule

3

Schedule rule

Runs every

5

Minutes

Rules run periodically and detect alerts within the specified time frame.

Additional look-back time

15

Minutes

Adds time to the look-back period to prevent missed alerts.

We then put the rule with our name and family name

2 About rule

Name
CR450-Frederic-Perron

Description
450

Default severity
Select a severity level for all alerts generated by this rule.
☒ Low

☐ Severity override
Use source event values to override the default severity.

Default risk score
Select a risk score for all alerts generated by this rule.
0 25 50 75 100 21

☐ Risk score override
Use a source event value to override the default risk score.

Tags Optional
Type one or more custom identifying tags for this rule. Press enter after each tag to begin a new one.

We duplicate the rule

Duplicate the rule with exceptions?

You are duplicating 1 selected rule, please select how you would like to duplicate the existing exceptions

☒ Duplicate rule and their exceptions
☐ Only duplicate the rule

Cancel Duplicate

We then add the winlog string after running the command prompt test.

event.code : "4625" and event.outcome : "failure" and user.name : * and winlog.event_data.TargetU

Group by
host.ip x

Threshold
1

Select fields to group by. Fields are joined together with 'AND'
Count
winlog.event_data.TargetUserName x

Unique values
3

Select a field to check cardinality
Timeline template
None

Select which timeline to use when investigating generated alerts.

We can then run random tests. We simply chose the names hello1, hello2, and hello3 for the users.

```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.19041.1415]
(c) Microsoft Corporation. All rights reserved.

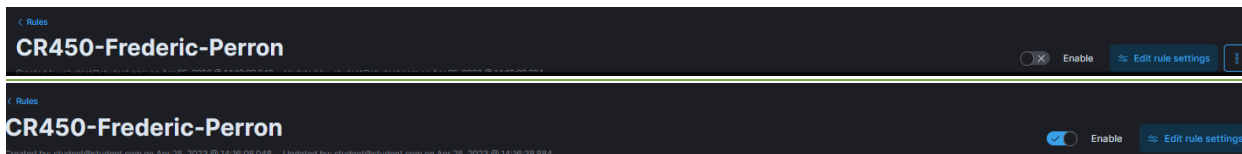
C:\WINDOWS\system32>runas /user:hello1 cmd
Enter the password for hello1:
Attempting to start cmd as user "ADHD\hello1" ...
RUNAS ERROR: Unable to run - cmd
1326: The user name or password is incorrect.

C:\WINDOWS\system32>runas /user:hello2 cmd
Enter the password for hello2:
Attempting to start cmd as user "ADHD\hello2" ...
RUNAS ERROR: Unable to run - cmd
1326: The user name or password is incorrect.

C:\WINDOWS\system32>runas /user:hello3 cmd
Enter the password for hello3:
Attempting to start cmd as user "ADHD\hello3" ...
RUNAS ERROR: Unable to run - cmd
1326: The user name or password is incorrect.

C:\WINDOWS\system32>_
```

Here we can see the rule triggered after 5 minutes



- b) Enable two rules of your choice from the built-in rules in Elastic. Use any tool of your choice (Atomic, PowerShell, etc.) to trigger the rules. Document your results (20p)

For each, include a screenshot:

- Of the rule
- Of the base event (used to trigger the rule)
- Of the correlated event

We start by adding the proxy rule.

```
C:\WINDOWS\system32>netsh interface portproxy show all

Listen on ipv4:          Connect to ipv4:
Address      Port      Address      Port
-----
127.0.0.1    8080      192.168.182.158 80
```

For example, I still do not see an alert, only a warning.

Port Forwarding Rule Addition

Created by: student@student.com on Apr 25, 2023 @ 14:00:08.772 Updated by: student@student.com on Apr 25, 2023 @ 19:13:38.727

Last response: ● warning at Apr 25, 2023 @ 19:18:43.002

Warning at Apr 25, 2023 @ 19:18:43.002

This rule is attempting to query data from Elasticsearch indices listed in the "Index pattern" section of the rule definition, however no index matching: ["winlogbeat-*"] until a matching index is created or this rule is disabled.

This is the rule we will use along with the information and commands.

RDP Enabled via Registry

Created by: student@student.com on Apr 25, 2023 @ 14:00:08.780 Updated by: student@student.com on Apr 25, 2023 @ 14:00:08.780

Last response: ● --

About

Identifies registry write modifications to enable Remote Desktop Protocol (RDP) access. This could be indicative of adversary lateral movement preparation.

Author	Elastic
Severity	● Medium
Risk score	47
License	Elastic License v2
MITRE ATT&CK™	Lateral Movement (TA0008) 🔗 - Remote Services (T1021) - Remote Desktop Protocol (T1021.001)
Timestamp override	event.ingested

Definition

Index patterns

logs-endpoint.events* winlogbeat* logs-windows*
endgame*

Custom query

```
registry where host.os.type == "windows" and event.type in ("creation", "change") and registry.path : ( "HKLM\\SYSTEM\\CurrentControlSet\\Control\\Terminal Server\\fDenyTSConnections", "REGISTRY\\MACHINE\\SYSTEM\\CurrentControlSet\\Control\\Terminal Server\\fDenyTSConnections" ) and registry.data.strings : ("0", "0x00000000") and not (process.name : "svchost.exe" and user.domain == "NT AUTHORITY\\") and not process.executable : "C:\\Windows\\System32\\SystemPropertiesRemote.exe"
```

Rule type

Event Correlation

We then execute the command in PowerShell.

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Try the new cross-platform PowerShell https://aka.ms/powershell

PS C:\WINDOWS\system32> Set-ItemProperty -Path 'HKLM:\System\CurrentControlSet\Control\Terminal Server' -name "fDenyTSConnections" -Value 0
```

We then restart the services so the machine now accepts RDP.

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\WINDOWS\system32> Restart-Service TermService -Force
```

Before attempting to trigger the alert, we activate the rule and wait a few minutes as recommended.

The screenshot displays the Elastic SIEM interface for a rule titled "RDP Enabled via Registry". The rule is currently disabled, as indicated by the "Enable" toggle switch. The "About" section on the left provides details about the rule, including its author (Elastic), severity (Medium), risk score (47), and license (Elastic License v2). The "Definition" section on the right shows the index patterns and the custom query used by the rule. The index patterns are "logs-endpoint.events.*", "winlogbeat-*", "logs-windows.*", and "endgame-*". The custom query is a KQL query that filters for registry write modifications to enable Remote Desktop Protocol (RDP) access.

RDP Enabled via Registry Enable [Edit rule settings](#)

Created by: student@student.com on Apr 25, 2023 @ 14:00:08.780 Updated by: student@student.com on Apr 25, 2023 @ 19:28:00.810
Last response: [View logs](#)

About [Details](#) [Investigation guide](#) [Setup guide](#)

Identifies registry write modifications to enable Remote Desktop Protocol (RDP) access. This could be indicative of adversary lateral movement preparation.

Author Elastic

Severity Medium

Risk score 47

License Elastic License v2

Definition

Index patterns logs-endpoint.events.* winlogbeat-* logs-windows.* endgame-*

Custom query registry where host.os.type == "windows" and event.type in ("creation", "change") and registry path in ("HKLM\\SYSTEM\\(*ControlSet*\\Control\\Terminal Server\\)\\DenyTSConnections", "(\\REGISTRY\\MACHINE\\SYSTEM\\(*ControlSet*\\Control\\Terminal Server\\)\\DenyTSConnections") and

We can then see the alert displayed in our *Onion*.